1 Linear Equations

Let $F = \mathbb{R}$ or $\mathbb{C}$.

Definition 1.1 (Linear equation)**.** A *linear equation* in the variables $x_1, \cdots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots a_nx_n = b \tag{1.2}$$

where b and the *coefficients* $a_1, \cdots, a_n$ are elements of F, and the subscript n is a positive integer.

1. Determine whether each of the following equations is linear.

   (a) $4x_1 - 5x_2 + 2 = x_1$

   (b) $x_2 = 2\sqrt{x_1} - 6$

   (c) $2x_1 + (1+i)x_2 - x_3 = 2\sqrt{6}$

   (d) $4x_1 - \sqrt{5}x_2 = x_1x_2$

Definition 1.3 (System of linear equations)**.** A *system of linear equations* (or a *linear system*) is a collection of one or more linear equations involving the *same* variables $x_1, \cdots, x_n$.

Example 1.4. The following is a system of two linear equations in three variables x_1, x_2, x_3:

$$\begin{aligned} 2x_1 - x_2 + 1.5x_3 &= 8 \\ x_1 \qquad - \ 4x_3 &= -7 \end{aligned} \tag{1.5}$$

Definition 1.6 (Solution)**.** Any n-tuple $(s_1, \cdots, s_n)$ of numbers which satisfies *each* equation in a linear system when $s_1, \cdots, s_n$ are substituted for $x_1, \cdots, x_n$ is called a *solution* of the system.

Example 1.7 (Testing a solution)**.** The 3-tuple $(5, 6.5, 3)$ is a solution of the system (1.5) since

$$\begin{aligned} 2(5) - 6.5 + 1.5(3) &= 8 \\ 5 \qquad - \ 4(3) &= -7 \end{aligned}$$

Definition 1.8 (Solution set)**.** The set of all solutions is called the *solution set* of the linear system.

Definition 1.9 (Consistent and inconsistent systems)**.** If a system of equations has at least one solution it is said to be *consistent*. If the system has no solutions, it is said to be *inconsistent*.

2. Is $(3, 4, -2)$ a solution of the following linear system?

$$\begin{aligned} 5x_1 - x_2 + 2x_3 &= 7 \\ -2x_1 + 6x_2 + 9x_3 &= 0 \\ -7x_1 + 5x_2 - 3x_3 &= -7 \end{aligned}$$

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Example 1.10 (Two linear equations in two unknowns)**.** Take $F = \mathbb{R}$ and consider the most general linear system of two equations in two unknowns

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 &= b_1, \\ A_{21}x_1 + A_{22}x_2 &= b_2. \end{aligned} \tag{1.11}$$

Assuming A_{12} and A_{22} are nonzero, we can rewrite each equation in slope-intercept form, so that the system (1.11) becomes

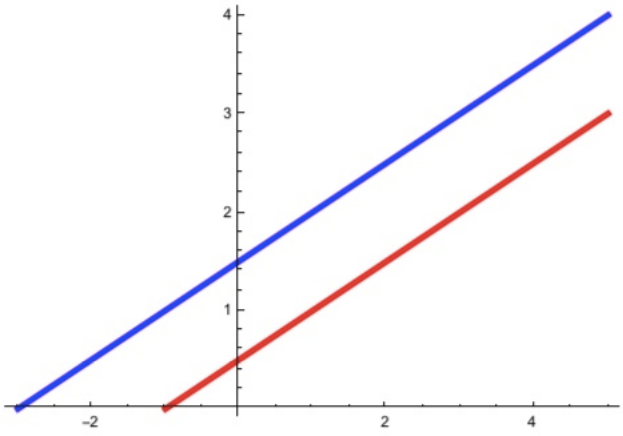
$$\begin{aligned} x_2 &= -\frac{A_{11}}{A_{12}}x_1 + \frac{b_1}{A_{12}}, \\ x_2 &= -\frac{A_{21}}{A_{22}}x_1 + \frac{b_2}{A_{22}}. \end{aligned} \tag{1.12}$$

Geometrically, each equation describes a line in the plane. Since these equations are in the same variables, the two lines lie in the same plane. A solution to the linear system corresponds to a point which lies on *both* lines at the same time, i.e., it is a point of intersection of the two lines. It is geometrically evident that there are three possibilities for the possible solution sets of the linear system (1.11), depending on the coefficients:

(1) The linear system (1.11) has *no solution* when

$$\begin{aligned} \frac{A_{11}}{A_{12}} &= \frac{A_{21}}{A_{22}}, \\ b_1 &\neq b_2, \end{aligned}$$

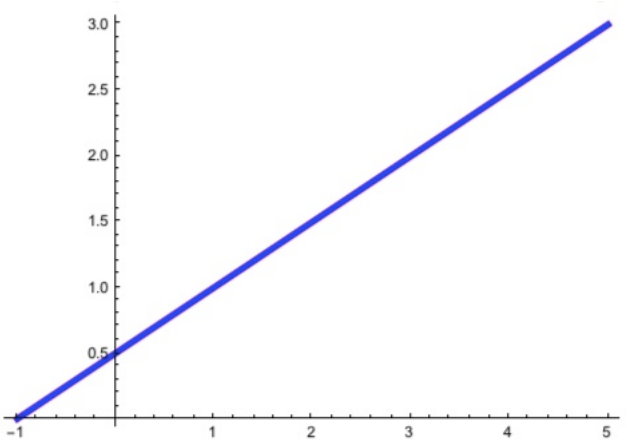
i.e., when the lines are parallel (same slope) but non-overlapping (different y-intercepts).

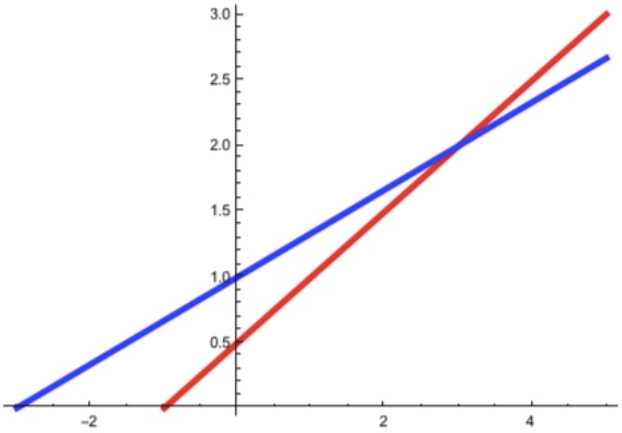


(2) The linear system (1.11) has *infinitely many solutions* when

$$\begin{aligned} \frac{A_{11}}{A_{12}} &= \frac{A_{21}}{A_{22}}, \\ b_1 &= b_2, \end{aligned}$$

i.e., when the lines are overlapping (both described by exactly the same equation).





(3) The linear system (1.11) has a *unique solution* in all other cases (i.e., when the lines are non-parallel).

In general, we will consider a linear system of m equations in n unknowns:

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &= y_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &= y_2 \\ \vdots \qquad\quad & \vdots \qquad\qquad \vdots \quad \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &= y_m \end{aligned} \tag{1.13}$$

where $y_1, \cdots, y_m$ and $A_{ij}, 1 \leq i \leq m, 1 \leq j \leq n$ are given. In (1.13), we denote the coefficient of x_j in the ith equation as A_{ij}.

We will prove that the three possibilities in Example 1.10 are the only possibilities in general; that is, the general linear system in (1.13) has either

1. No solution,
2. A unique solution, or
3. Infinitely many solutions.

Example 1.14. We can again understand this claim geometrically in the case of three linear equations in three unknowns:

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + A_{13}x_3 &= y_1 \\ A_{21}x_1 + A_{22}x_2 + A_{23}x_3 &= y_2 \\ A_{31}x_1 + A_{32}x_2 + A_{33}x_3 &= y_3. \end{aligned}$$

These are the point-normal equations of three planes in $\mathbb{R}^3$. The possible intersections are shown below, giving rise to the same three possibilites:

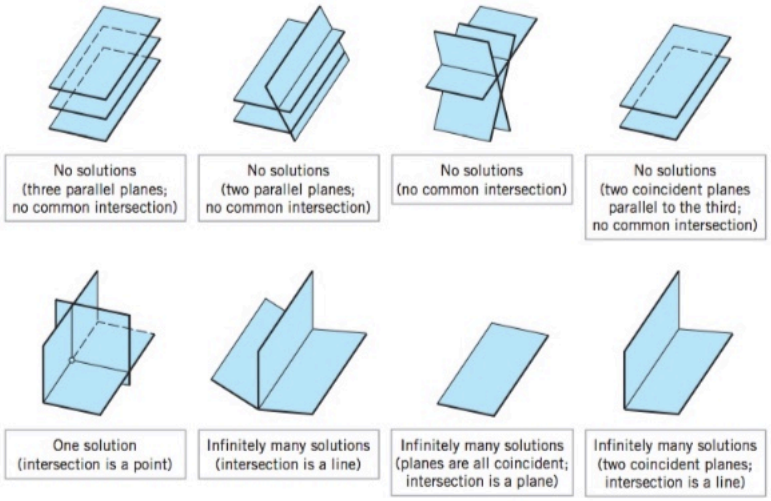


Figure 1: The intersections of three planes give the solution sets of a linear system of three equations in three unknowns.

Definition 1.15 (Homogeneous and inhomogeneous sytems)**.** If $y_1 = y_2 = \cdots = y_m = 0$ in (1.11), the system is said to be *homogeneous*. Otherwise, it is said to be *inhomogeneous*.

3. A homogeneous system always has at least one solution. Can you find it?